Numerical processing of fin. data – Session 5

6/02/2026

Exercises

Reminders:

- The Black-Scholes PDE from the lectures

$$\begin{cases} \partial_t P(t,s) + r s \partial_s P(t,s) + \frac{1}{2}\sigma^2 s^2 \partial_{ss} P(t,s) - r P(t,s) = 0 \\ P(T,s) = g(s) \end{cases} \qquad (\text{PDE}_{BS})$$

- We are given the representation:

$$P(t,s) = E\left[\underbrace{e^{-r(T-t)}}_{\text{discount factor}} g\left(s\, e^{\left(r-\frac{\sigma^2}{2}\right)(T-t) + \sigma\sqrt{T-t}\, G}\right)\right] \qquad \boxed{\forall s>0, \forall t \in (0,T)}$$

- Note that

$$\left(r-\frac{\sigma^2}{2}\right)(T-t) + \sigma\sqrt{T-t}\, G \overset{\text{law}}{=} N\left(\left(r-\frac{\sigma^2}{2}\right)(T-t), \sigma^2(T-t)\right)$$

Indeed, since $G \sim N(0,1)$

we have $a + bG \sim N(a, b^2)$

Exercise 1 | Call option: $g(S_T) = (S_T - K)^+$ for some $K > 0$

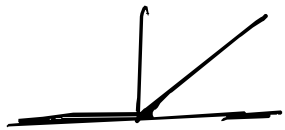
1) Starting from Equation (2):

$$p^{Call}(t,s) = E\left[e^{-r(T-t)}(Y-K)^+\right] = (*)$$

where $Y = s\, e^{(r-\frac{\sigma^2}{2})(T-t) + \sigma\sqrt{T-t}\, G}$

here : $s, r, \sigma, T, t \rightarrow$ constants

$G \rightarrow$ random



$$\overset{*}{=} e^{-r(T-t)} E\left[(Y-K)^+\right] = (*)$$

We can use the fact that

$$(Y-K)^+ = (Y-K)\, 1_{Y-K \geq 0}$$

$$= (Y-K)\, 1_{Y \geq K}$$

where $1_{Y \geq K} = \begin{cases} 1 & \text{if } Y \geq K \\ 0 & \text{otherwise} \end{cases}$

$$1_{\{Y \geq K\}}(\omega) = \begin{cases} 1 & \text{if } \omega \in \{Y \geq K\} \\ 0 & \text{if } \omega \notin \{Y \geq K\} \end{cases}$$

$\omega \in \Omega$

$$(*) = e^{-r(T-t)}\left(E\left[Y 1_{Y \geq K}\right] - K\, E\left[1_{Y \geq K}\right]\right) \checkmark$$

2) We work on the term $E[1_{Y \geq K}]$

We have:

$$E[1_{Y \geq K}] = \mathbb{P}(Y \geq K)$$ ← rewrite this probability in terms of G

$$= \mathbb{P}\left(S e^{(r-\frac{\sigma^2}{2})(T-t) + \sigma\sqrt{T-t}\, G} \geq K\right)$$

$$= \mathbb{P}\left(\left(r-\frac{\sigma^2}{2}\right)(T-t) + \sigma\sqrt{T-t}\, G \geq \ln\left(\frac{K}{S}\right)\right)$$

$$= \mathbb{P}\left(\sigma\cancel{\sqrt{T-t}}\, G \geq \frac{1}{\sigma\sqrt{T-t}}\left(\ln\left(\frac{K}{S}\right) - \left(r-\frac{\sigma^2}{2}\right)(T-t)\right)\right)$$

$$= \mathbb{P}\left(G \geq \frac{1}{\sigma\sqrt{T-t}}\left[\ln\left(\frac{K}{S}\right) - \ln e^{r(T-t)}\right] + \frac{\sigma}{2}\sqrt{T-t}\right)$$

$$= \mathbb{P}\left(G \leq \frac{1}{\sigma\sqrt{T-t}} \ln\left(\frac{S e^{r(T-t)}}{K}\right) - \frac{1}{2}\sigma\sqrt{T-t}\right)$$ by symmetry

$$= \boxed{\mathbb{P}(G \leq d_2(t,S))} = \boxed{N(d_2(t,S))}$$

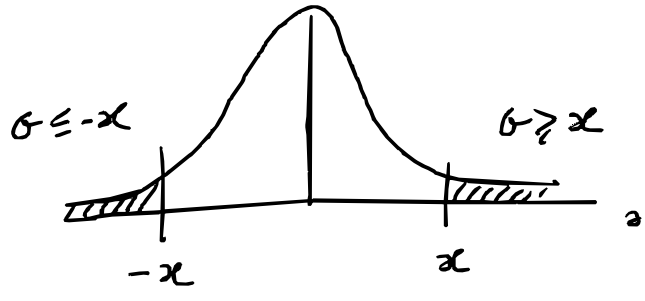
We have:

$$\mathbb{P}(f(G) \geq K) = \mathbb{P}(G \geq f^{-1}(K))$$

if f invertible, and increasing

<u>Symmetry</u> of $N(0,1)$:

$$\mathbb{P}(G \geq x) = \mathbb{P}(G \leq -x), \quad \forall x \in \mathbb{R}$$



3) We work on the term:

$$E[Y \mathbb{1}_{Y \geq K}] = E\left[S e^{(r-\frac{\sigma^2}{2})(T-t) + \sigma\sqrt{T-t}\, G} \mathbb{1}_{\{Y \geq K\}} \right]$$

$$\overset{(**)}{=} S e^{r(T-t) - \frac{\sigma^2}{2}(T-t)} E\left[e^{\sigma\sqrt{T-t}\, G} \mathbb{1}_{\{G \geq -d_2(t,s)\}} \right]$$

because we have just seen that

$$\{Y \geq K\} = \{ G \geq -d_2(t,s) \}$$

$$\overset{(**)}{=} S e^{r(T-t)} e^{-\frac{\sigma^2}{2}(T-t)} \int_{-d_2(t,s)}^{+\infty} e^{\sigma\sqrt{T-t}\, x} e^{-\frac{x^2}{2}} \frac{dx}{\sqrt{2\pi}}$$

density of G

We can notice that:

$$-\frac{x^2}{2} + \sigma\sqrt{T-t}\, x - \frac{1}{2}\sigma^2(T-t) = -\frac{1}{2}\left(x - \sigma\sqrt{T-t}\right)^2 \quad \forall x$$

$$\overset{(**)}{=} S e^{r(T-t)} \int_{-d_2}^{+\infty} e^{-\frac{1}{2}(x - \sigma\sqrt{T-t})^2} \frac{dx}{\sqrt{2\pi}}$$

change of variable

$$= S e^{r(T-t)} \int_{-d_2 - \sigma\sqrt{T-t}}^{+\infty} e^{-\frac{1}{2}y^2} \frac{dy}{\sqrt{2\pi}}$$

$y = x - \sigma\sqrt{T-t}$

$\boxed{d_2 = d_2(t,s)}$

$$= S e^{r(T-t)} \mathbb{P}\left(G \geq -d_2 - \sigma\sqrt{T-t} \right)$$

$$= S e^{r(T-t)} \mathbb{P}\left(G \leq \underbrace{d_2 + \sigma\sqrt{T-t}}_{d_1} \right)$$

by symmetry

$$= S e^{r(T-t)} N(d_1(t,s)) \checkmark$$

Conclusion: we have shown that the Black Scholes formula for a call option price is

$$\boxed{p^{Call}(t,s) = S\, N(d_1(t,s)) - e^{-r(T-t)} K\, N(d_2(t,s))}$$

Rmk: d_1 and d_2 are explicit functions

the whole formula is explicit (as soon as we can evaluate the Gaussian cdf $N(\cdot)$)

<u>Exercise 2</u>: Goal: find the formula for the Black-Scholes delta

$$\boxed{\delta_t = \partial_s P^{call}(t,s)}$$

1) Start from noticing a certain identity:

$$S\, n(d_1) = K e^{-r(T-t)} n(d_2) \qquad \forall S \quad \forall K \quad \forall t, \forall T$$

where $n(x) = \dfrac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}$

In the identity : $\rightarrow$ LHS $= S\,n(d_1)$

$\rightarrow$ RHS $= K e^{-r(T-t)}\, n(d_2)$

$$\boxed{\sqrt{2\pi}\, S\, n(d_1)} = S \exp\left(-\frac{d_1^2}{2}\right)$$

$$= S \exp\left(-\frac{1}{2}\left(d_2 + \sigma\sqrt{T-t}\right)^2\right) \quad \text{since } d_2 = d_1 - \sigma\sqrt{T-t}$$

$$= S \exp\left(-\frac{1}{2}d_2^2 - \frac{1}{2}\sigma^2(T-t) - \sigma\sqrt{T-t}\, d_2\right)$$

$$= \exp\left(-\frac{1}{2}d_2^2\right) \exp\left(\cancel{-\frac{1}{2}\sigma^2(T-t)}\right) S \exp\left(-\ln\left(\frac{S}{K}\right) - r(T-t) + \cancel{\frac{1}{2}\sigma^2(T-t)}\right)$$

$$= \sqrt{2\pi}\, n(d_2)\, \underbrace{S \exp\left(-\ln\frac{S}{K}\right)}_{S\cdot\frac{K}{S} = K} e^{-r(T-t)}$$

by definition of d_2

$$= \boxed{\sqrt{2\pi}\, n(d_2)\, K e^{-r(T-t)}} = \text{RHS} \quad \checkmark$$

2) <u>Delta of the BS call:</u>

$$\partial_s P^{call}(t,s)$$

$$= \frac{d}{ds}\Big(\underbrace{S\,N(d_1(t,s)) - Ke^{-r(T-t)}N(d_2(t,s))}_{\text{depend on } s}\Big)$$

$$= N(d_1) \boxed{+S\,n(d_1)\,\partial_s d_1(t,s) - Ke^{-r(T-t)}\,n(d_2)\,\partial_s d_2}$$

↳ Is this term $=0$?

We have used $N'(x) = n(x)$

Note that

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

$$\partial_s d_2 = \partial_s d_1$$

Therefore, the term in the box is:

$$\partial_s d_1 \Big(\underbrace{S\,n(d_1) - Ke^{-r(T-t)}\,n(d_2)}_{=0 \to \text{ from } q1}\Big)$$

Conclusion

$$\delta^{call}_{BS}(t,s) = \boxed{\partial_s P^{call}(t,s) = N(d_1(t,s))} \in [0,1]$$

<u>Notice:</u> The delta of a call option is between 0 and 1